

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☒ ☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☒ ☐ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☒ ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☐ ☒ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software used
Data analysis	Radiocarbon dates Bayesian analysis of radiocarbon dates was performed in OxCal v. 4.4 (https://c14.arch.ox.ac.uk/oxcal.html) ZooMS Mass spectrometry identifications were performed using the open-source software MMass (http://www.mmass.org/) Cementum analysis Measurements of the increments were conducted using ImageJ (https://imagej.nih.gov/ij). PCR amplification Sequences were visually inspected and manually curated, assembled and aligned using Geneious software suite v9.1.8 (https://www.geneious.com). Library construction and sequencing Sequence authenticity was estimated on libraries subjected to either a partial or no USER treatment using MapDamage. We drew alleles at random using PileupCaller (https://github.com/stschiff/sequenceTools), rendering the individuals from the dataset homozygous for each locus. The two files were then merged using Plink v1.9. We then conducted principal component analysis using smartPCA implemented in EIGENSOFT. We mapped our filtered reads to both the Ovis aries genome assembly (international Sheep Genome consortium build Oarv3.1) and the Capra hircus genome assembly (ARS1) using BWA. We produced a multiple genome alignment of our newly reconstructed sequences with Geneious software suite v9.1.8. Gaps were removed from the alignment and a Maximum Parsimony tree was built using MEGA version X. Phylogenetic trees were visualized and manipulated in FIGTREE v1.4.1 (http://tree.bio.ed.ac.uk/software/figtree/).

Graphs

All graphs were produced and edited in R (<https://cran.r-project.org/>) and Adobe Photoshop

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Sequencing data can be found on the European Nucleotide Archive: [to be added upon upload]

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description

This study is a mixed-methods, interdisciplinary analysis of archaeological materials from the site of Obishir V, in Kyrgyzstan. All available archaeological materials (as of 2017) were selected for the analysis, of which the best preserved specimens (n=6) were selected for genomic sequencing.

Research sample

All available skeletal remains from 2017 and early excavations were chosen for study. For ZooMS analysis, all materials from one square of the excavation (R8) were selected for study.

Sampling strategy

All available materials from the archaeological assemblage were chosen for study.

Data collection

Researchers were blind to any hypothesis during data collection, which did not involve human participants.

Excavation

Excavations at Obishir V site carried out between 2015–2019 by a joint Russian-Kyrgyzstan archaeological expedition. In 2015–2016, researchers working at the site excavated an area of 8 m² adjoining the 1968–1969 excavation area (Supplementary Appendix A). Each artifact (lithic and faunal remains) greater than 1 cm in size was individually piece-plotted with 1 mm accuracy using a Leica Total Station TS02 plus in tandem with the Trimble software EDM Mobile, and documented along with contextual information (orientation, dip, inclination, find category, stratigraphic level, etc.). Fragments smaller than this were collected in bulk for each 0.25 m² area. Features (natural or anthropogenic) were recorded with the total station using outlines, surface points and breaklines. Large surfaces, such as walls of old excavation pits, sections and archaeological horizons with numerous artefacts were also documented using photogrammetry and 3D-scanning. Digital records were supplemented by field drawings and notes. All sediment obtained during the excavation was wet-screened with mesh of a size of 2.0 mm, and finds were documented according to excavated square, layer and depth. Stratigraphic units were defined during the excavation based on grain size composition, consistency, color, presence of erosional surfaces and accumulations of limestone clasts.

Thermoluminescence dating

During the 2017 field season, two sediment samples were collected from the southern portion of the western excavation profile of layer 5 at a depth of 295 and 335 cm below the surface. We measured deposit moisture in each sample, and after drying determined dose rate (DR) using a MAZAR gamma spectrometer. Concentrations of ²²⁶Ra, ²²⁸Th, ⁴⁰K in each sample were obtained from twenty measurements lasting 2000 s each. We established equivalent dose (ED) on the 63–80 mm polymineral fraction, after 10% HCl and 30% H₂O₂ washing and UV optical bleaching. The samples were irradiated with 20 Gy, 30 Gy, 40 Gy, 50 Gy and 100 Gy, doses from ⁶⁰Co gamma source. Before measurement, we heated the samples at 140 degrees for 3 hours. A sample pre-treated in this way was used to determine the equivalent dose (ED) by the TL multiple-aliquot regenerative technique (67), according to the description published by Fedorowicz et al. (68). Curve registration was performed on RA'94 (Mikrolab) thermoluminescence reader, coupled with EMI 9789 QA photomultiplier. Finally, we calculated TL age according to Frechen (69).

Zooarchaeology and Zooarchaeology by Mass Spectrometry

For each specimen, we conducted taphonomic analysis and morphology-based species identifications using comparative faunal collections held by the Russian Academy of Sciences-Siberia in their field station in Aidarkyen, Kyrgyzstan, and attempted refits for all specimens to control for issues of fragmentation. We recorded taphonomic indicators such as rodent and carnivore damage, root etching, and weathering (70), along with evidence for anthropogenic modification such as spiral fracturing, cut marks, and burning via visual inspection (Supplementary Appendix C). In addition, we used collagen fingerprinting (or Zooarchaeology by Mass Spectrometry, also known as ZooMS) to taxonomically identify a subset of this material, all derived from 2017 excavations. We

analyzed all bone specimens from Square R8, Layers 2.2, 3, and 4, as well as all bone specimens from Square R8, Layer 1.3 using ZooMS. To do so, we demineralized 10 mg of bone using 50mM ammonium bicarbonate buffer (Sigma-Aldrich) produced in UHQ water and pH adjusted to 8.0 using Ammonium Hydroxide (1M), following the protocol outlined by (71). The extracted collagen was digested into component peptides using trypsin (Pierce), and was purified using Pierce C18 tips (Thermo Scientific), eluting in 50% acetonitrile (Sigma-Aldrich) and 0.1% TFA (Sigma-Aldrich). Samples were then spotted on Bruker AnchorChip with Bruker Peptide Calibration Standard in calibration spots directly neighbouring the samples. Mass spectrometric analysis was conducted using a Bruker Autoflex Speed LRF MALDI-TOF in the dedicated ZooMS Laboratory at the Max Planck Institute for the Science of Human History in Jena, Germany.

Cementum analysis

First, all analyzed teeth were 3D scanned using a MicroCT scanner, and washed in a 90% ethanol solution. A cast of the occlusal surface was retained for later micro-wear analyses. Thin sections were prepared following recommendations by Rendu (73). Teeth were embedded in epoxy within a vacuum chamber (at 0.2 bars pressure). They were then cut at a thickness of 250µm using an Isomet5000 automatic saw, and ground to a thickness of 100µm with a PressiCube grinding machine. For each tooth, four independent thin sections were prepared and analyzed independently. Observations were conducted under polarized light and polarized/analyzed light both with and without the insertion of a lambda plate (74), using a Leica 2500P microscope equipped with a Leica DC 120 camera. P

DNA Extraction

DNA was extracted from 6 teeth using silica-based purification in a dedicated laboratory for ancient DNA at PACEA (UMR 5199, University of Bordeaux). We incubated between 100-250 mg of bone powder in a 1 mL extraction buffer (0.5 M EDTA, 0.25 M Na2HPO4, 50 µg/mL proteinase K) for 48-72 h at 37°C. DNA was purified from the collected supernatant using a method adapted from the QIAquick MinElute purification kit protocol (Qiagen).

PCR amplification

Purified DNA was amplified by qPCR using a LightCycler® 96 Instrument (Roche Life Science). Primer pairs were designed to amplify 2 regions of the mitochondrial DNA, a 130 bp region from cytB and 170bp from the hypervariable region (Supplementary Appendix F: list of primers used and product sizes). We obtained each PCR product at least twice, conducting standard negative control to ensure authenticity of the results. We used 1 µL of extract in 10 µL reaction with 1X FastStart Essential DNA Green Master and 1µM of each primer. Products were sequenced by capillary electrophoresis at Genewiz (Leipzig, Germany).

Library construction and sequencing

We used DNA extracts to build double-stranded Illumina libraries following the protocol described in Gorge et al (77). We sequenced whole genome shotgun libraries for ~ 500 000 reads on an Illumina NextSeq 500 at the Institut de Recherche Biomédicale des Armées in Brétigny-sur-Orge, France, to assess DNA preservation. DNA extracts of five samples produced in Bordeaux were sent to the dedicated clean room facilities of the Max Planck Institute for the Science of Human History. There, 20 µl of the extract were used to build double stranded UDG-half libraries (78). The resulting genetic libraries were double indexed and further amplified (79). Finally, all five samples were shotgun sequenced for ~20M reads on an Illumina HiSeq run with a 75-cycles single-end configuration.

Timing	Archaeological research was conducted in the summer of 2017, with ZooMS conducted in Fall 2017, and radiocarbon dating and DNA sequencing from early 2018 through spring of 2020.
Data exclusions	No data were excluded from the analyses.
Non-participation	No human participants were used in the study
Randomization	No human participants were used in the study, and no data were allocated into groups

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☐	☒ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Palaeontology and Archaeology

Specimen provenance	Specimens all come from the archaeological site of Obishir V in Aidaryken, Kyrgyzstan, and research was conducted under Permit #0040/02-05 issued by the The Field Committee of the Institute of History and Cultural Heritage of the National Academy of Sciences, Kyrgyz Republic on 19 April, 2017. Specific provenance for each object is provided in the supplementary material.
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Specimen deposition

All specimens are stored in the collections at the Institute for Archaeology and Ethnography, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russian Federation

Dating methods

Nine radiocarbon samples on identified and unidentified animal bone and tooth samples ($n = 7$) and charcoal ($n = 2$) were selected from various depths across the stratigraphic unit of layers 2-4, and submitted for radiocarbon dating at the Oxford Radiocarbon Accelerator Laboratory, the Center for Isotope Research at the University of Groningen, and the Golden Valley Laboratory at Novosibirsk. A portion of tooth 20-1 was submitted to the University of Arizona Accelerator Mass Spectrometry Laboratory. Collagen was obtained from 20-1 dentin using Acid-Base-Acid (ABA) pretreatment, gelatinization, 0.45 micron filtration, and ultrafiltration. Quality control parameters: collagen yield (7.1%); carbon yield (42.9%); CN ratio (3.2); and $\delta^{13}C$ (19.7 ± 0.1 per mil), indicated good preservation (64). All dates were calibrated in OxCal using the INTCAL13 calibration curve (65). We performed Bayesian stratigraphic phase modeling (using ordered phases and combining Layers 2-4) in the software OxCal with a uniform prior using all available radiocarbon dates from Obishir V, and two thermoluminescence dates from Layer 5 (using OxCal's C_date function), following methods outlined by Ramsey (66). We also modeled the cultural occupation of Layer 2 separately, using only bone and tooth samples using a uniform prior. The models provided good agreement, and repeating the analysis using a general outlier model did not significantly alter estimated parameters. OxCal code used in the analysis, along with human tooth dating methods are outlined in Supplementary Appendix B.

☒ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

All research was conducted in concurrence with ethical guidelines of the Max Planck Institute for the Science of Human History, Jena, Germany.

Note that full information on the approval of the study protocol must also be provided in the manuscript.